FRE-GT-9743 Assignment 4
Question 1

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Question 1: Why OTC prices carry a discount factor but futures prices do not

In both markets cash moves every day. The difference is *what* that cash is.

OTC (collateralised under a CSA): the cash is a deposit, not a settlement

When I post collateral equal to $V(t) - C(t)$, I am not paying for the trade. I am handing over a security deposit. The contract itself is untouched: it still promises its payoff at maturity T. The collateral sits with my counterparty, earns interest (the collateral rate), and is returned when the trade ends.

Because the real payoff is still sitting at time T, the value today must convert that future cash flow into today's money:

$$V(t) = \mathbb{E}^{\mathbb{Q}}[\, D(t,T)\,\text{Payoff}(T) \mid \mathcal{F}_t\,].$$

The collateral only removes credit risk. It does not change the timing of the payoff, so the discount factor stays.

Futures: the cash *is* the settlement

Variation margin is not a deposit. It is final payment of the day's gain or loss:

$$F(t_{k+1},T) - F(t_k,T).$$

Once it is paid, the money is gone for good. Nobody returns it later, and no interest is paid on it. The contract then resets to zero value. So at any date there is no large payoff waiting at T. The only cash flow ahead is the next day's price change.

The discount factor for that one day, $D(t_k, t_{k+1})$, is already known at t_k. It is just a positive constant multiplying a quantity whose expectation must be zero (entry is free), so it cancels:

$$0 = D(t_k,t_{k+1})\Big(\mathbb{E}^{\mathbb{Q}}\big[F(t_{k+1},T) \mid \mathcal{F}_{t_k}\big] - F(t_k,T)\Big) \implies F(t_k,T) = \mathbb{E}^{\mathbb{Q}}\big[F(t_{k+1},T) \mid \mathcal{F}_{t_k}\big].$$

Chaining these one-day steps and using $F(T,T) = S(T)$ gives

$$F(t,T) = \mathbb{E}^{\mathbb{Q}}\big[S(T) \mid \mathcal{F}_t\big],$$

with no discounting.

Another way to see it

The futures price $F(t,T)$ is not the *value* of anything. The value of a futures position is always zero after settlement. F is a *quoted level*, like a strike that resets every day, and you only need to discount values. An OTC price $V(t)$, by contrast, is the value of a claim on future money, so it must be discounted.

Summary

- **OTC:** collateral is a returnable, interest-bearing deposit. The payoff still occurs at T, so $V(t)$ includes $D(t,T)$.
- **Futures:** margin is a final, non-returnable daily settlement. Each step looks only one period ahead, its known discount factor cancels, and $F(t,T)$ is a pure (undiscounted) $\mathbb{Q}$-expectation.